

# GHZ Numeric Notes

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## I. INTRODUCTION

### A. Corrector Tensor

The corrector tensor  $x_{\mu\nu}$  is determined by the source  $T_{ab}$  through a hierarchy of equations descending from the  $l^a$ -sector (IRG variant) of Einstein's equations  $l^a(\mathcal{E}_{ab}[x] = T_{ab})$ . In GHP form, these equations read [?] ]

$$\rho^2 \mathbb{P} \left( \frac{\bar{\rho}}{\rho^3} \mathbb{P} \left[ \frac{\rho}{\bar{\rho}} x_{m\bar{m}} \right] \right) = T_{ll} \quad (\text{I.1})$$

for  $x_{m\bar{m}}$ ,

$$\begin{aligned} & \frac{\rho}{2(\rho + \bar{\rho})} \mathbb{P} \left( (\rho + \bar{\rho})^2 \mathbb{P} \frac{x_{nm}}{\rho(\rho + \bar{\rho})} \right) \\ &= T_{lm} - \frac{1}{2} \{ (\mathbb{P} + \rho - \bar{\rho})(\delta + \bar{\tau}' - \tau) + 2\bar{\tau}'(\mathbb{P} - 2\rho) - (\delta - \tau - \bar{\tau}')\bar{\rho} + 2\rho\tau \} x_{m\bar{m}}. \end{aligned} \quad (\text{I.2})$$

for  $x_{nm}$ , and

$$\begin{aligned} & \frac{1}{2}(\rho + \bar{\rho})^2 \mathbb{P} \left( \frac{1}{\rho + \bar{\rho}} x_{nn} \right) \\ &= T_{ln} - \text{Re} \{ \delta' \delta + (\bar{\tau}' - \tau) \delta' + [\delta(\bar{\tau} + \tau')] - 2\tau' \tau - 2\Psi_2 - 2\bar{\rho} \rho' + 2(\rho' \mathbb{P} + \rho \mathbb{P}') - \mathbb{P}' \mathbb{P} \} x_{m\bar{m}} \\ & \quad - \text{Re} \{ [(\mathbb{P} - 2\rho)(\delta' - \bar{\tau}) + (\tau' + \bar{\tau})(\mathbb{P} + \bar{\rho}) - 2(\delta' - \tau')\rho - 2\bar{\tau} \mathbb{P}] x_{nm} \} \end{aligned} \quad (\text{I.3})$$

for  $x_{nn}$ .

In this note, we give explicit solutions to equations (I.1), (I.2), and (I.3) for a point particle (IV.1) in terms of readily computable coefficients written in terms of the source.

We work in mostly outgoing Kerr-Newman coordinates wherein equations (I.1), (I.2), and (I.3) reduce to a sequence of three ODEs along the outgoing null-generators  $l^\alpha = \partial_r$  of increasing  $r$  the affine parameter labeling cuts of  $\mathcal{I}^+$ , or in BL coordinates. For initial data, we assume trivial conditions on  $x_{\alpha\beta}$  at  $\mathcal{H}^-$ , consistent with a bound orbit supported away from the horizon for all time.

Throughout the note, we adopt the mostly minus metric “west coast” signature  $(+, -, -, -)$  and geometrized units  $G = c = 1$ . Greek letters denote spacetime indices, lowercase Latin letters spatial indices, capital Latin letters denote quantities on the  $\mathbb{S}^2$ .

To avoid tracking factors of  $\pi$ , we adopt the convention wherein the perturbed EFE reads

$$\mathcal{E}_{\mu\nu} = T_{\mu\nu}. \quad (\text{I.4})$$

Unless stated otherwise, all other conventions and notation will be chosen to match GHZ.

## II. PRELIMINARIES

### III. CALCULATION OF $x_{\mu\nu}$

#### A. General solution in the source-free region

The solution to (I.1) in the vacuum region  $r > r_{\max}$  is given by

$$x_{m\bar{m}}^> = a^\circ \left( \frac{\bar{\rho}}{\rho} + \frac{\rho}{\bar{\rho}} \right) + b^\circ (\rho + \bar{\rho}), \quad r > r_{\max}, \quad (\text{III.1})$$

where  $\rho/\bar{\rho} + \bar{\rho}/\rho$  and  $\rho + \bar{\rho}$  are homogeneous solutions. The integration constants  $a^\circ$  and  $b^\circ$  are determined by the source and will be given in a later section.

We next turn to the  $nm$ -component, as determined by (I.2). To integrate this equation, we write it first in Held form

$$\frac{\rho}{2(\rho + \bar{\rho})} \mathbb{P} \left( (\rho + \bar{\rho})^2 \mathbb{P} \frac{x_{nm}}{\rho(\rho + \bar{\rho})} \right) = T_{lm} - \frac{1}{2} \left\{ \bar{\rho}(\mathbb{P} + \rho - \bar{\rho})\tilde{\delta} - \bar{\rho}(3\bar{\rho} + \rho)\tau^\circ \mathbb{P} + 4\rho\bar{\rho}^2\tau^\circ \right\} x_{m\bar{m}}, \quad (\text{III.2})$$

wherein we have used Eqs. (??),(??), and (??). Integrating the result give

$$x_{nm}^> = \bar{\rho}\Omega^\circ\tilde{\delta}a^\circ + \frac{(\rho + \bar{\rho})^2}{\rho}\tau^\circ a^\circ + \bar{\rho}\tilde{\delta}b^\circ + \bar{\rho}(\rho + \bar{\rho})\tau^\circ b^\circ \\ + \frac{1 - \Omega^{\circ 2}\bar{\rho}^2}{\bar{\rho}}c^\circ + \rho(\rho + \bar{\rho})e^\circ, \quad r > r_{\max}. \quad (\text{III.3})$$

We begin the calculation of  $x_{nn}$  by recasting Eq. (I.3) in Held form

$$\frac{1}{2}(\rho + \bar{\rho})^2\mathbf{P} \left( \frac{x_{nn}}{\rho + \bar{\rho}} \right) = T_{ln} + \text{Re}\mathcal{U}x_{m\bar{m}} + \text{Re}[\mathcal{V}x_{nm}], \quad (\text{III.4})$$

where

$$\mathcal{U} = \rho\bar{\rho} \left[ -\tilde{\delta}'\tilde{\delta} - \bar{\tau}^\circ(\rho + \bar{\rho})\tilde{\delta} + \bar{\tau}^\circ\tilde{\delta}\mathbf{P} + \tau^\circ(\bar{\rho} - \rho)\tilde{\delta}' + \tau^\circ\tilde{\delta}'\mathbf{P} \right] + \tilde{\mathbf{P}}'\mathbf{P} \\ + \frac{1}{2} \left[ \Psi^\circ(\rho^2 - \bar{\rho}^2) - 2\rho\bar{\rho}(\bar{\rho} - \rho)\tau^\circ\bar{\tau}^\circ - 4\bar{\rho}\rho'^\circ(1 + \rho\Omega^\circ) \right] \mathbf{P} - 2\rho\tilde{\mathbf{P}}' \\ + \rho^2\Psi^\circ(2\rho - \bar{\rho}) + 2\bar{\rho}^2(\rho'^\circ - \rho(2\rho - \bar{\rho})\tau^\circ\bar{\tau}^\circ) - \rho\bar{\rho}(\rho - 3\bar{\rho})\rho'^\circ\Omega^\circ, \quad (\text{III.5})$$

and

$$\mathcal{V} = -\rho\{-3\rho\tilde{\delta}' + [\tilde{\delta}' - \bar{\tau}^\circ(2\bar{\rho} + \rho)]\mathbf{P} + 2\rho^2\bar{\tau}^\circ\} \quad (\text{III.6})$$

The solution is given by

$$x_{nn}^> = 2\text{Re} \left\{ \frac{1}{2}\frac{\rho + \bar{\rho}}{\rho^2}\tilde{\mathbf{P}}'a^\circ + (1 + \rho\bar{\rho}\Omega^{\circ 2})\tilde{\delta}'\tilde{\delta}a^\circ + 2(\rho^2 + 2\rho\bar{\rho})\tau^\circ\bar{\tau}^\circ a^\circ \right. \\ + \left( \frac{\rho^2}{\bar{\rho}} + \frac{3}{2}\rho\bar{\rho}\Omega^\circ \right) \Psi^\circ a^\circ - 2\rho'^\circ a^\circ + \tilde{\mathbf{P}}'b^\circ + 2\rho\bar{\rho}\bar{\tau}^\circ\tilde{\delta}b^\circ + 2\rho^2\bar{\rho}\tau^\circ\bar{\tau}^\circ b^\circ + \rho^2\Psi^\circ b^\circ \\ \left. + \rho^2\tilde{\delta}'e^\circ + 2\rho^2\bar{\rho}\bar{\tau}^\circ e^\circ - \frac{\bar{\rho}}{\rho} \left( 4\Omega^\circ + \frac{1}{\rho} \right) \tilde{\delta}'c^\circ + 4\frac{\bar{\rho}}{\rho}\bar{\tau}^\circ c^\circ + (\rho + \bar{\rho})f^\circ \right\} \quad (\text{III.7})$$

#### IV. POINT-PARTICLE SOURCE

Let us now take the perturbation to be sourced by a particle of mass  $\mu$  on a worldline  $\gamma$  parametrized by proper time here denoted  $\lambda$ . We further denote the position and four velocity of the particle by  $x_p^\mu(\lambda)$  and  $v^\mu = dx_p^\mu/d\lambda$ , respectively. It carries the point particle stress energy distribution

$$T_{\mu\nu} = \mu \int_\gamma v_\mu v_\nu \delta_4(x, x_p(\lambda)) d\lambda. \quad (\text{IV.1})$$

We consider a bound orbit with radii between  $r_{\min} < r_p < r_{\max}$  and denote the coordinate location of the particle in retarded coordinates as  $x_p^\mu = (u_p, r_p, z_p, \varphi_p)$ .

### A. Held Decomposition of the source

With retarded KN time  $-\infty < u_p < \infty$  taken as the time parameter, we now rewrite the source as

$$T_{\mu\nu} = \int_{-\infty}^{\infty} \left[ \mu \frac{v_\mu v_\nu}{\Sigma_p v^u} \delta_p^2(\Omega) \right] \delta_p(u) \delta_p(r) du_p, \quad (\text{IV.2})$$

$$= T_{\mu\nu}^\circ(u, z, \varphi; r_p) \delta_p(r), \quad (\text{IV.3})$$

$$= J_{\mu\nu}^\circ(u; r_p, z_p, \varphi_p) \delta_p^2(\Omega) \delta_p(r), \quad (\text{IV.4})$$

where in general  $x^p(u)$  after doing the  $u_p$  integration, and where (in the outgoing Kerr chart)

$$T_{\mu\nu}^\circ := \mu \frac{v_\mu v_\nu}{\Sigma_p \gamma_p} \delta_p^2(\Omega), \quad J_{\mu\nu}^\circ := \mu \frac{\gamma_p \dot{x}_\mu \dot{x}_\nu}{\Sigma_p}, \quad (\text{IV.5})$$

with the coordinate four-velocity

$$\dot{x}^\mu := \frac{dx^\mu}{d\lambda} \frac{d\lambda}{du} = \frac{v^\mu}{\gamma_p} \quad (\text{IV.6})$$

and  $\delta_p^2(\Omega) = \delta(z - z_p) \delta(\varphi - \varphi_p)$  and where  $v^\alpha \partial_\alpha u = \gamma_p$ . Here,  $J_{\mu\nu}^\circ$  and  $T_{\mu\nu}^\circ$  are taken at  $r_p(u)$ ,  $\Sigma_p = r_p^2 + a^2 z_p^2$  and it is understood that  $r_p = r_p(\lambda(u_p))$  etc. We also introduce the quantities  $T_\mu^\circ$  and  $J_\mu^\circ$  by

$$T_{\mu\nu} l^\nu = T_\mu^\circ(u, z, \varphi) \delta_p(r) = J_\mu^\circ(u; z_p, \varphi_p) \delta_p(r) \delta_p^2(\Omega) \quad (\text{IV.7})$$

## V. SPECTRAL SOLVER

## VI. WINDOWING THE PUNCTURE